RR interval features and a lightweight CNN for sleep apnea classification

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Abstract

This retrospective study examined whether RR interval variability improves sleep apnea classification and how feature-based classifiers compare with a lightweight waveform convolutional neural network (CNN). The analysis included 34 PhysioNet Apnea-ECG recordings comprising 16,558 one-minute windows. Models were evaluated using five outer grouped folds, with hyperparameter and threshold selection confined to inner folds. Adding variability features to basic interval and heart-rate summaries increased mean outer-fold average precision (AP) by 0.107-0.248 across three classical classifiers. Interval standard deviation had the largest conditional contribution. In the tuned comparison, logistic regression achieved the highest observed mean AP (0.646), followed by an adapted literature linear SVM (0.603); the 100 Hz CNN achieved 0.474. However, performance varied substantially across folds, and all primary models had pooled specificity below 0.41 at their selected thresholds. These findings support the predictive value of RR variability within this experiment but show that useful ranking performance did not produce reliable operating points. Unverified subject identities, prior cohort exposure and the absence of external validation limit the conclusions to exploratory internal evaluation.

1. Introduction

A feature-based ECG classifier depends on what its input summaries preserve. Mean interval and heart-rate measures describe cardiac level, while variability measures describe changes between beats. A waveform model instead learns from the filtered samples. Comparing these approaches on the same retained windows makes it possible to examine the information added by engineered variability features and the performance of a small learned representation.

The study used two research questions to separate feature effectiveness from pipeline comparison. Its purpose was to evaluate a defined set of representations and models under a common grouped protocol, including the cases where ranking and thresholded decisions gave different impressions of performance.

The task was to classify existing one-minute annotations using ECG alone. It was not to diagnose a person, estimate a clinical apnea-hypopnea index, or replace respiratory measurements.

RQ1 - Feature effectiveness: How do different ECG-derived feature representations affect sleep apnea classification performance?

A basic representation, F1, contained mean RR, median RR and mean instantaneous heart rate. The expanded representation, F2, added SDRR, RR-based RMSSD and pRR50. Removing each added feature in turn tested its contribution conditional on the remaining inputs.

RQ2 - Model comparison: How do traditional feature-based models compare with a lightweight deep learning model for sleep apnea detection?

Logistic regression, random forest and XGBoost using F2 were compared with a compact 1D CNN using filtered ECG waveforms. An independently implemented adaptation of a published linear SVM method

provided a literature-derived comparator. This was a comparison of complete input-and-model pipelines: it did not isolate classifier choice from representation.

2. Methods

2.1 Data source and study cohort

The source was the PhysioNet Apnea-ECG database, version 1.0.0. Its ECG is sampled at 100 Hz, and the reference annotations were assigned by experts using simultaneously recorded respiratory and related signals. The database documents that `c05` overlaps `c06`; `c05` was excluded. Only the learning cohort was analyzed. The separate challenge test cohort was not used as external validation.

Table 1. Cohort characteristics.

Cohort characteristic	Value
Retained recordings	34
Literature-derived groups	24
Complete one-minute windows	16,558
Apnea-associated windows	6,508 (39.3%)
Non-apnea windows	10,050 (60.7%)
Windows with at least one quality flag	623 (3.76%)
Windows with missing F1/F2 predictors	115 (0.69%)

Each window contained 6,000 samples. Source A/N labels at the corresponding minute boundary were retained; the positive label includes apnea/hypopnea annotations. These labels should not be interpreted as a precise event-onset or event-duration target.

The grouping used a published record-to-subject mapping, documented in Notebook 01 and attributed to Yeh et al. [2]. Original subject identifiers were unavailable. Group separation was enforced computationally, but verified patient independence cannot be claimed.

2.2 Filtering, beats and quality policy

Source files were checked against recorded checksums. ECG was filtered within each full recording using a 0.5-40 Hz Butterworth bandpass, design order four, in second-order sections with forward/backward zero-phase filtering. This is retrospective processing: it uses future samples within a recording and is not a causal streaming pipeline.

R peaks were detected with WFDB GQRS in blocks with surrounding context, then locally aligned and deduplicated. Detected RR intervals were accepted between 0.3 and 2.0 seconds. Features used intervals whose endpoints lay within the same minute. Differences between successive intervals preserved original adjacency; rejecting an interval did not join its neighbors. No ectopic-beat classification or RR interpolation was performed.

Complete finite windows were retained even when flagged for flat signal, low amplitude, too few usable intervals or suspect RR intervals. Flags were diagnostic metadata, not predictors. Missing features were

imputed within the fitting partition. This policy kept the cohort identical across representations and avoided silently restricting the comparison to cleaner signals.

2.3 Feature definitions

Table 2. Feature representations and definitions.

Representation	Feature	Definition
F1	Mean RR	Mean accepted interval, in ms
F1	Median RR	Median accepted interval, in ms
F1	Mean instantaneous heart rate	Mean of 60,000 divided by each accepted RR interval, in bpm
F2 addition	SDRR	Sample standard deviation of accepted RR intervals, in ms
F2 addition	RR-based RMSSD	Root mean square of valid adjacent interval differences, in ms
F2 addition	pRR50	Percentage of valid adjacent differences with absolute magnitude greater than 50 ms

At least 20 accepted intervals were required; RMSSD and pRR50 also required ten valid adjacent pairs. Undefined values remained missing until fold-local imputation. These are detected RR statistics, not audited normal-to-normal measurements. Their predictive value does not validate a physiological interpretation of one-minute HRV.

2.4 Grouped nested evaluation

Five outer folds held each literature-derived group out once. Within each outer training partition, three inner grouped folds were shared across models. Imputation, scaling and fitting-dependent class weights used only the current training partition. Identifiers, groups, quality flags and labels were excluded from predictors.

The two questions used different controls:

- **RQ1:** classifier settings were fixed across all five feature configurations. Each model and its preprocessing were refitted for every configuration. This kept feature effects separate from hyperparameter changes.

- **RQ2:** each of the five pipelines received 24 frozen candidate configurations per outer fold. Logistic regression and the linear SVM used grids; tree models used seeded sampled configurations. The CNN architecture was fixed, with its training settings searched. Selection maximized mean inner-fold AP.

For every selected pipeline, a threshold maximized F1-score on inner out-of-fold predictions, with the lowest threshold breaking ties. That threshold was then applied unchanged to the outer held-out groups. Outer labels did not select hyperparameters, epochs or thresholds. The evaluation protocol, candidate lists, input hashes and source versions were frozen before these outer evaluations.

2.5 Models and CNN training

Classical models used median imputation; logistic regression and the SVM additionally used standard scaling. The literature comparator adapted Almazaydeh et al. [3] using 11 local RR statistics and a linear SVM. Its windows were 60 seconds, its beat extraction followed this project's common policy, and recording-wide summaries were omitted. It was an adaptation, not an exact reproduction or execution of author code. Expert reference labels were not a human-performance baseline.

The CNN had 3,009 trainable parameters: three convolutional blocks, ReLU activations, two max-pooling layers, global average pooling and a linear output. Inputs were filtered 100 Hz minutes, standardized within each minute using a 0.001 standard-deviation floor, clipped to ±8, and stored as float32. Training used Adam and binary cross-entropy with logits; inference applied sigmoid once.

CNN epochs were calibrated using a stopping subset drawn only from inner training groups. Unweighted stopping BCE used patience eight, minimum improvement 0.0001 and a maximum of 60 epochs. Each candidate was then freshly initialized and refitted on all inner-training groups for its selected epoch count before inner validation. The outer refit used the rounded median of the selected inner epoch counts. This separated stopping feedback from both inner validation and outer evaluation.

Primary results used seed 67. Random forest, XGBoost and CNN were also refitted with seeds 167 and 267, retaining selected settings and epoch schedules. These runs measured refit sensitivity, not independent cohort replications; no seed was selected as the winner and predictions were not ensembled.

2.6 Metrics and uncertainty

The primary outcome was the **arithmetic mean of five outer-fold AP scores**, using scikit-learn's average-precision definition. AP summarizes precision across recall levels; it is not classification accuracy. A constant scorer's mean fold AP was 0.395, reflecting mean fold prevalence.

Pooled outer predictions supplied secondary AP, ROC-AUC, F1-score, sensitivity and specificity. Pooled scores combine outputs from five fitted models, so pooled AP need not equal mean fold AP. Thresholded metrics used each fold's own inner-selected threshold.

Whole groups were resampled jointly across methods for 2,000 paired bootstrap draws. The reported percentile intervals describe differences in **pooled AP conditional on the fitted predictions**. They do not cover all training or tuning uncertainty and are not intervals for the primary mean fold AP. No minute-level significance test was used. Fold standard deviations are descriptive.

2.7 Post hoc implementation and signal checks

After evaluation, independent checks reconstructed the cached CNN inputs, replayed saved model predictions and recalculated inner-selected thresholds. Selected calibration histories were checked against the stopping rule. Near-flat intervals were traced to original WFDB digital and physical samples, source annotations and whole-record filtering. These were exploratory diagnostics; they did not change the evaluated models, thresholds or cohort.

3. Results

3.1 RR feature ablations

Expanded versus basic features.

Table 3. Basic and expanded features under fixed classifier settings.

Fixed-setting classifier	Basic F1 AP	Expanded F2 AP	F2 - F1
Logistic regression	0.401	0.650	+0.248
Random forest	0.400	0.507	+0.107
XGBoost	0.390	0.531	+0.141

Values are means across matched outer folds; differences are calculated before rounding. The basic summaries alone were close to the constant-score reference. Adding the three variability features improved mean AP for all classifiers. The gain occurred in all five folds for logistic regression and four of five for each tree model. It was not uniform across groups or models.

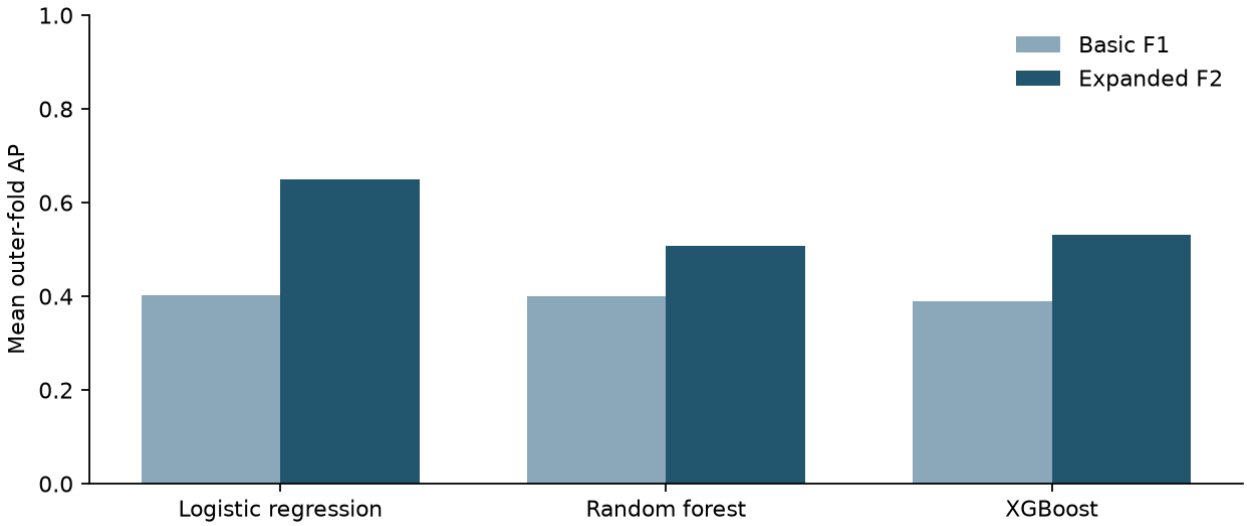


Figure 1. RR variability improved mean AP under fixed classifier settings.

Conditional feature removal.

Table 4. Conditional feature-removal results.

Fixed-setting classifier	Full F2	Without SDRR	Without RMSSD	Without pRR50
Logistic regression	0.650	0.357	0.521	0.668
Random forest	0.507	0.355	0.489	0.519
XGBoost	0.531	0.363	0.503	0.545

Removing SDRR reduced AP by 0.293, 0.152 and 0.168, respectively. Removing RMSSD also reduced mean AP, although its effect was smaller in the tree models. Both removals reduced AP in every fold for all three classifiers.

Removing pRR50 modestly increased mean AP for all three models, by 0.012-0.018. The fold-level changes were mixed, so this is evidence of limited incremental value in this representation, not a consistent benefit from removing it everywhere. The prespecified F2 representation remained the basis of RQ2.

3.2 Tuned pipeline comparison

Mean performance.

Table 5. Tuned pipeline performance.

Tuned pipeline	Mean fold AP	Fold SD	Pooled AP	Pooled ROC-AUC
Logistic regression, F2	**0.646**	0.137	**0.594**	**0.708**
Adapted literature linear SVM	0.603	0.171	0.560	0.684
XGBoost, F2	0.540	0.136	0.508	0.630
Random forest, F2	0.502	0.098	0.499	0.633
Lightweight CNN, 100 Hz	0.474	0.101	0.432	0.548
Constant-score reference	0.395	0.053	0.393	0.500

Logistic regression exceeded the CNN by 0.172 mean AP and XGBoost by 0.106. The adapted SVM was a substantial competitor, exceeding both tree models and the CNN. Its score should not be compared directly with the source paper's headline accuracy because the cohort, windows and evaluation differ.

The bounded searches did not produce consistent gains over fixed F2 settings: AP changed from 0.650 to 0.646 for logistic regression, 0.507 to 0.502 for random forest, and 0.531 to 0.540 for XGBoost. Better inner selection does not guarantee better held-out performance. Both analyses are retained rather than choosing whichever scored better on the outer data.

Fold and seed variability.

Table 6. Average precision by outer fold.

Pipeline	Fold 1 AP	Fold 2 AP	Fold 3 AP	Fold 4 AP	Fold 5 AP
Logistic regression	0.662	0.476	0.856	0.614	0.623
Adapted linear SVM	0.633	0.438	0.849	0.651	0.444
XGBoost	0.633	0.518	0.720	0.437	0.392
Random forest	0.459	0.535	0.657	0.448	0.408
CNN	0.366	0.486	0.406	0.631	0.483

The CNN led fold 4 among the three F2 classical models, despite its lower overall mean. Logistic regression ranged from 0.476 to 0.856 AP. Fold prevalences also differed, from 0.319 to 0.468, which matters when interpreting AP. The spread is not solely an estimate of algorithm randomness, and overlapping training folds are not independent experiments.

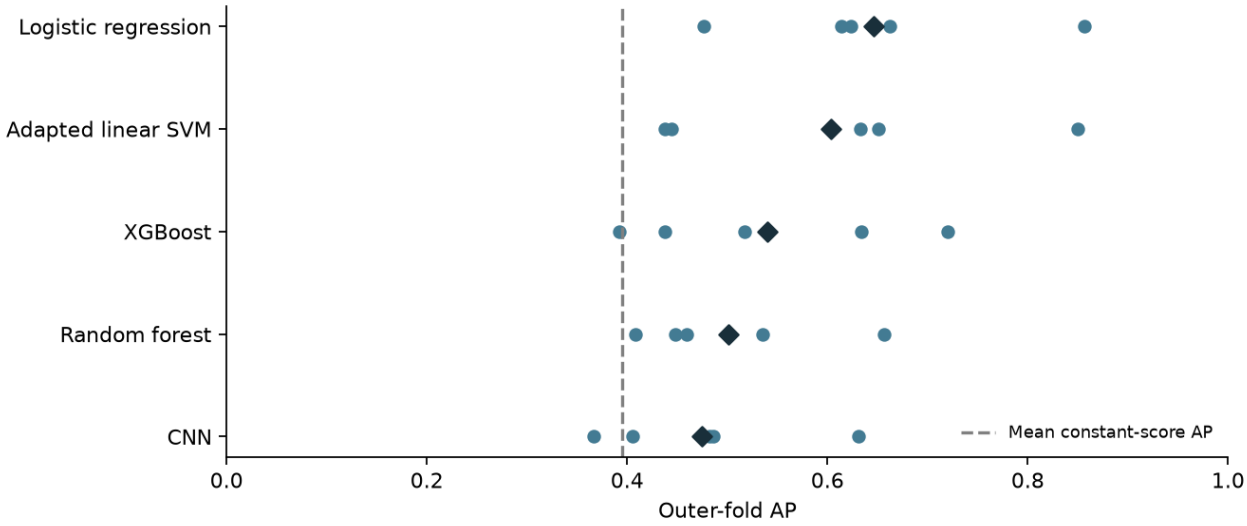


Figure 2. Five fold scores and their means; dashed line: constant-score reference.

Repeat-seed mean AP was 0.502/0.503/0.502 for random forest, 0.540/0.540/0.541 for XGBoost and 0.474/0.487/0.476 for the CNN. Seed changes did not reverse the main ordering among these pipelines, but small mean seed variation does not resolve uncertainty about unseen groups.

Conditional uncertainty.

Table 7. Conditional paired bootstrap intervals for pooled AP differences.

Pooled AP difference: XGBoost minus comparator	Estimate	Conditional 95% interval
Logistic regression	-0.086	-0.164 to -0.008
Adapted linear SVM	-0.052	-0.133 to +0.037
Random forest	+0.009	-0.023 to +0.063
CNN	+0.076	-0.074 to +0.236

All comparisons had 2,000 valid draws. The LR comparison offers conditional exploratory evidence favoring LR over XGBoost. The other intervals cross zero. These intervals are not adjusted for multiple comparisons, and direct LR-CNN or LR-SVM intervals were not part of the exported comparison set. The mean ranking should not be described as proof that every adjacent pair differs reliably.

3.3 Thresholded performance

Table 8. Pooled performance at the inner-selected operating points.

Pipeline	Pooled sensitivity	Pooled specificity	Pooled F1-score
Logistic regression	0.760	0.406	0.568
Adapted linear SVM	0.766	0.398	0.569
XGBoost	0.763	0.309	0.539
Random forest	0.743	0.347	0.540
CNN	0.712	0.345	0.523
Always predict apnea	1.000	0.000	0.564

The always-positive reference is important: its pooled F1-score was close to those of logistic regression and the SVM, and above those of the other models. Useful ranking signal therefore did not translate into a persuasive operating point. The selected models exchanged many false alarms for sensitivity.

The CNN predicted every held-out minute positive in folds 4 and 5. Random forest and XGBoost also did so in fold 1; XGBoost specificity in fold 2 was only 0.0004. A plausible-looking F1-score can conceal zero specificity.

The threshold audit showed that the problem began before outer evaluation:

Table 9. CNN positive prediction rates before and after outer refitting.

CNN fold	Inner-selected threshold	Inner positive rate	Outer positive rate
1	0.425	59.5%	75.8%
2	0.287	50.8%	40.7%
3	0.369	51.7%	26.7%
4	0.247	99.98%	100%
5	0.240	85.5%	100%

In fold 4, inner median scores for negative and positive minutes were 0.378 and 0.374, showing little useful separation. Maximizing F1 on those scores already favored calling almost everything positive. F1 maximization imposed no minimum specificity.

Other folds showed score-scale changes: in fold 1, the negative median rose from 0.378 on inner out-of-fold predictions to 0.629 on outer predictions. The observed pattern includes weak score separation, permissive threshold selection and changes in score distributions. These diagnostics did not isolate a single cause.

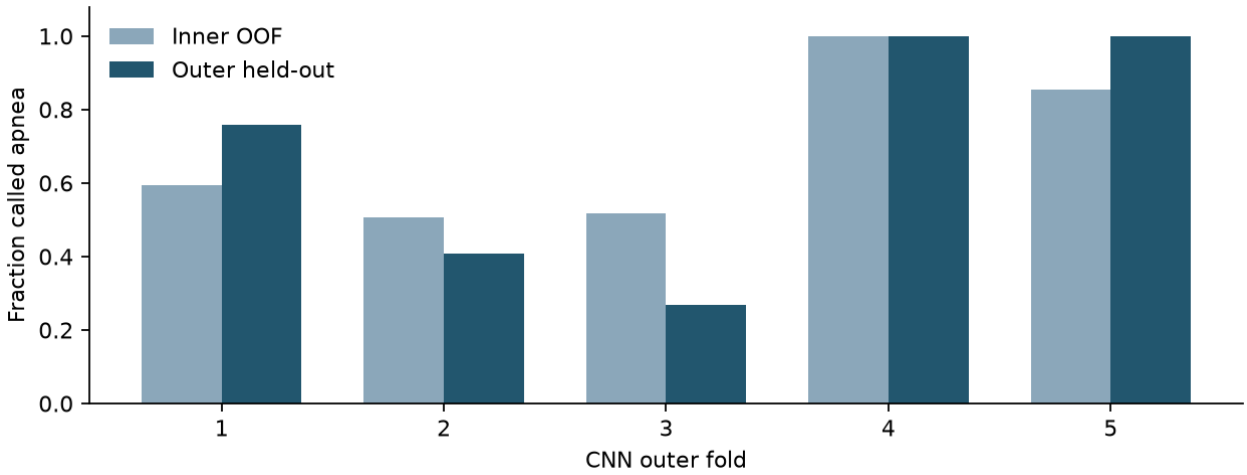


Figure 3. CNN inner thresholds already favored many positive calls.

3.4 Post hoc CNN and signal audit

Implementation checks.

All 16,558 cached CNN minutes matched independently reconstructed source preprocessing exactly. All 15 exported CNNs-five folds and three seeds-reproduced their saved outer scores with zero maximum absolute difference. Independent threshold calculations agreed with the saved primary thresholds across all five pipelines. These checks found no cache-ordering or inference-conversion error; they do not prove that every upstream decision was appropriate.

Selected training curves.

Seven of the 15 selected CNN calibration runs chose epoch 1. Other runs selected as late as epoch 23; the final outer schedules were 1, 1, 3, 2 and 1 epochs. The stopping decisions and refit lengths matched the specified rule.

Training loss decreased in every selected calibration run, while final stopping loss exceeded its selected minimum. For example, in fold 2 / inner 1, training BCE fell from 0.573 to 0.391 as stopping BCE rose from 0.758 to 2.106. Another inner split in the same fold improved until epoch 12. This is consistent with poor generalization to some stopping groups, not a patience-counter error.

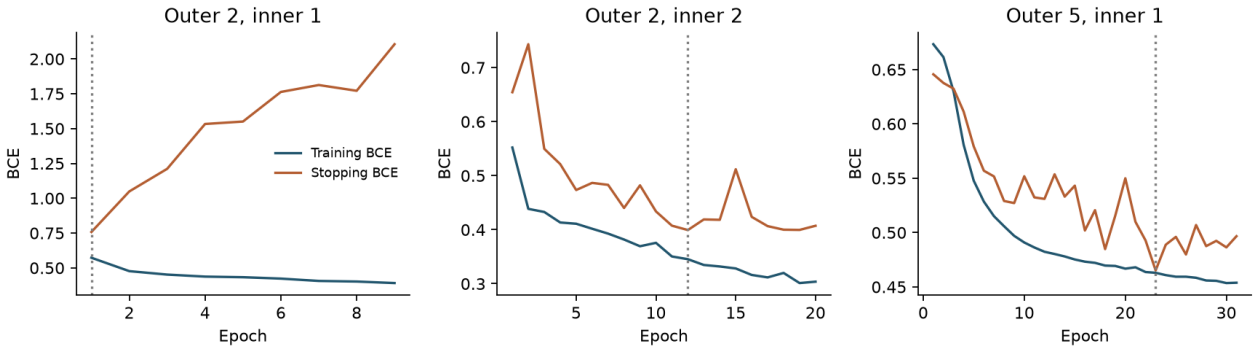


Figure 4. Selected training and stopping losses; dotted lines mark chosen epochs.

Source signal quality.

Seventeen filtered minutes fell below the normalization floor. Each contained exactly one digital value in the original WFDB recording: sixteen were constant at zero, and one at 2047 digital units. Saved raw physical samples matched original decoding. Independently reconstructed whole-record filtering matched the saved arrays exactly for all seven affected recordings, and the corresponding source annotations matched their retained labels.

All 17 minutes were already flagged as flat and low-amplitude. They were retained under the common quality policy, not introduced by CNN caching. Removing a constant baseline with a bandpass filter is expected behavior; these checks did not show that filtering erased valid ECG in those intervals. Their physical acquisition cause is unverified.

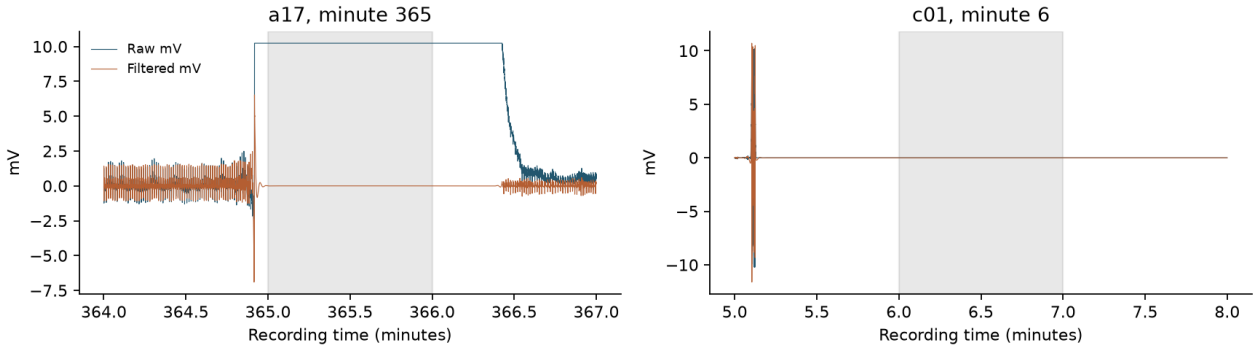


Figure 5. Shading marks constant-source minutes before and after filtering.

These minutes account for roughly 0.10% of the cohort. They establish a source-quality issue, but not an explanation for the overall performance gap. No windows were removed and no replacement scores were calculated after this finding.

4. Discussion

4.1 Interpretation of RQ1 and RQ2

Answer to RQ1: RR variability improved classification relative to basic interval and heart-rate summaries in this experiment. SDRR contributed most among the tested conditional removals; RMSSD added useful information, while pRR50 showed little additional benefit. These results describe predictive dependence within the tested feature sets, not physiological causation or universal feature importance.

Answer to RQ2: logistic regression achieved the highest observed mean AP among the tested pipelines, and the adapted literature SVM was also competitive. The compact CNN did not demonstrate an advantage. The evidence does not establish that classical models generally outperform deep learning, nor that this CNN result represents the best attainable waveform model.

The clearest finding is that interval variability mattered more than basic cardiac level summaries. Within the fixed ablation, SDRR and RMSSD added information that all three classifiers used. A six-feature linear model then led the tuned comparison. This favors a simple model as the starting point for further development on this task, rather than assuming a more complex estimator will generalize better.

The practical result is less favorable. Even the leading model combined sensitivity of 0.760 with specificity of 0.406. Fold differences, permissive thresholds and unstable CNN stopping histories show that the remaining problem is not adequately described by a single average score.

4.2 Training and operating-point implications

An inner threshold combines scores from several fitted models and may not match the scale of the final refit. The threshold audit therefore points to both permissive development operating points and score-distribution differences, rather than a single transfer failure.

These trajectories do not show that simply training longer would help. More observed epochs often worsened stopping BCE. BCE and AP measure different properties, but per-epoch stopping AP was not saved, so an objective mismatch remains a hypothesis. The median epoch rule also compresses heterogeneous histories into very short refits.

The fixed architecture offers another limitation: its local receptive field before global pooling spans 113 input samples, about 1.13 seconds at 100 Hz. It aggregates local responses across the minute without explicitly modeling long interval sequences. Whether that restriction explains its weak performance needs a separate experiment.

4.3 Limitations

Several limits govern interpretation:

1. **The independent sample is small and uncertain.** The 16,558 minutes are correlated within recordings. There are only 24 literature-derived groups, and original identities have not been verified. Grouped evaluation reduces a known form of overlap but cannot establish patient independence.

2. **This is exploratory internal evaluation.** The cohort had been examined before the protocol was frozen. Nested splitting controls selection within this run; it does not make the exposed cohort an untouched confirmatory test.

3. **There is no external validation.** Performance on another collection, acquisition setup or clinical population is unknown. Source annotations are a classification reference, not evidence of diagnostic utility.

4. **Signal and feature quality remain imperfect.** Constant intervals were present, beat detection was not audited against expert peaks throughout the cohort, and detected RR features are not verified NN measures. Retrospective zero-phase filtering also limits streaming claims.

5. **The model comparison has a defined scope.** It covers one compact CNN architecture, six-feature classical pipelines, an adapted local-RR SVM and finite searches. It does not isolate architecture from representation or establish a model-family ceiling. The SVM is not an exact reproduction or human baseline.

6. **Uncertainty is only partly measured.** Bootstrap intervals condition on saved predictions and omit full refitting/tuning variation. Repeated seeds add no independent subjects. Large fold variation remains relevant despite stable mean seed scores.

7. **Practical deployment was not assessed.** These thresholds were not selected for a specified clinical operating requirement, calibration was not established, and complete preprocessing-to-prediction serving cost was not benchmarked. Parameter count alone is not a deployment evaluation.

4.4 Further work

A useful next study would define signal-quality handling and operating-point requirements in advance, record per-epoch ranking metrics alongside loss, and evaluate a small set of training policies on development data. Any model changes prompted by these outer results remain exploratory on this cohort. Stronger evidence requires independent data with verified identities.

4.5 Conclusions

RQ1: adding RR variability improved mean AP across all three fixed-setting classifiers. SDRR had the strongest conditional contribution, RMSSD added information, and pRR50 provided little incremental value in the tested representation.

RQ2: logistic regression achieved the highest observed mean AP, followed by the adapted literature SVM. The compact 100 Hz CNN did not improve on the feature-based pipelines. The ranking does not establish universal model superiority, and the selected operating points were unreliable across held-out groups.

The study provides a reproducible comparison and a documented account of where the methods failed. Its strongest result is evidence for RR variability in this setting; its central limitation is uncertain generalization beyond the evaluated groups.

Data and code availability

The companion study repository contains the analyzed source recordings, feature tables, group assignments, reported predictions and metrics, notebook sources, executed evidence snapshots and 25 primary outer-fold models. Extra-seed and ablation predictions remain included; their model binaries and candidate checkpoints are omitted. Exact prepared waveforms are available as a separate checksum-verified release download.

The reproduction guide documents the environment, artifact manifests and execution order. Notebook 12 checks the report tables against saved results before rendering. Original code and associated documentation use the MIT license; PhysioNet data retain their separate ODC-By attribution terms. The full source and bundle manifests distinguish the completed evaluation from the curated distribution.

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